

JAYANTH DASAMANTHARAO

📍 New Jersey 📞 +1 (848)-313-5617 ✉ jayanthdasamantharao@gmail.com [🌐 LinkedIn](#) [🐙 Github](#) [📁 Portfolio](#)

PROFESSIONAL EXPERIENCE

AI Engineer - Fedway, EliteUS Software Solutions

Aug 2024 - Present | Piscataway, NJ

- Developed an AI chatbot utilizing **AWS Bedrock, LLMs, Azure** to facilitate system interactions and deliver human-like conversational experiences. Incorporated **Agentic AI, AutoGen** to enable multi-user functionality.
- Integrated **Azure Search Index, ChromaDB** vector databases and **LangChain**, to enhance interactions. Improved response accuracy through Retrieval-Augmented Generation **RAG** and deployed to **OneDev(GIT)**.
- Tech: Python, AWS Bedrock, Azure, LLMs, OpenAI, LangChain, AgenticAI, AutoGen, Prompt Engineering.

Data Science Intern, Harvest Software Solutions, LLC

June 2023 - Sept 2023 | Jacksonville, Florida

- Enhanced real-time ad campaigns by implementing **Machine Learning** models, resulting in a 70% improvement in predictive accuracy and efficiency. Performed **exploratory data analysis** on large-scale datasets.[5M+ rows]
- Collaborated with **cross-functional teams** to translate business requirements to data-driven solutions. Developed interactive dashboards in **Tableau**, effectively communicated strategic insights, business ideas to the client.

Database Analyst, Accenture Solutions Pvt. Ltd.

June 2021 - Aug 2022 | Hyderabad, India

Regeneron Pharmaceuticals Inc. - Client Data Analysis & Warehousing

- Optimized **ETL** workflows and implemented **automated data pipelines in Python**, enhancing data processing efficiency.
- Conducted large-scale data analysis using **SQL** and **AWS Athena, Redshift** while visualizing insights and built interactive dashboards in **Tableau** effectively communicating findings. Integrated **CI/CD pipelines** for faster deployments.
- Increased system reliability by 25% through **DDL optimization, seamless data integration, and performance tuning**, ensuring stable **data warehousing** and reporting operations using **SQL Workbench**.

Research Lab, Andhra University

January 2021 - June 2021 | Visakhapatnam, India

Self Driving Cars Using Image Processing and Deep Learning

- Engineered **Convolutional Neural Networks (CNNs)** for autonomous lane detection and object recognition using python.
- Leveraged **YOLO and Faster R-CNN** architectures to optimize real-time detection and classification accuracy.
- Enhanced model generalization through **image pre-processing** techniques like **Gaussian blurring and color space conversion**, and integrated **multi-sensor data** from **radar, lidar, and cameras** to improve navigational precision.
- Developed a real-time simulation environment using **Unity3D and a Flask backend** for dynamic parameter monitoring and system validation, enabling high-fidelity testing of autonomous driving functionalities.

TECHNICAL SKILLS

Programming: Python, R, Scala, MATLAB. **Libraries:** Pytorch, Scikit-learn, Tensorflow, SparkML, Tableau, PowerBI.
Machine Learning: Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Statistical Analysis: Causal Inference, Hypothesis testing, Time-series forecasting, stochastic process, Linear Regression.
Deep Learning: Neural Networks, Computer Vision, AI, NLP, BERT, GANs, LLMs, LLaMA, LSTM, OpenAI models.
Database & Cloud: SQL, AWS, RDBMS, Data Warehousing, OpenAI, Anthropic, Gemini, GCP, Langchain, Pinecone.
Language Models: OpenAI, Azure, AWS Bedrock, Hugging Face Transformers, LLaMA, prompt tuning, RAG, Langchain.

EDUCATION

Rutgers University, Masters in Data Science

Sept 2022 - May 2024 | New Brunswick, NJ

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** [Portfolio](#).

Andhra University, B.Tech in Electrical and Electronics Engineering

2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Bayesian Classification - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.

Thank you for considering my application. I'm available to start working immediately.